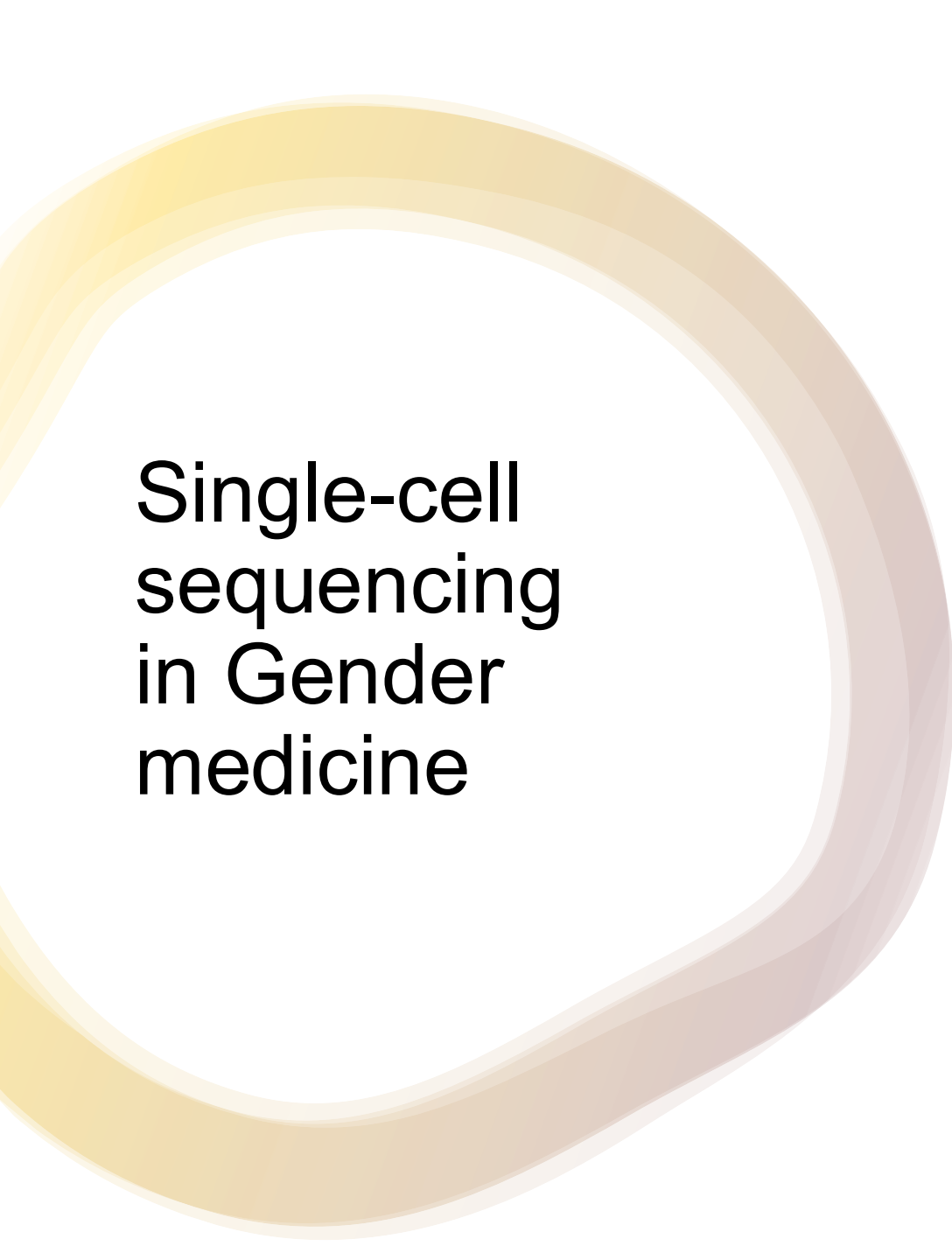




Single-cell sequencing in Gender medicine



Joana P. Bernardes

j.pimentabernardes@ikmb.uni-kiel.de

MOSAIC Summer school Day 3

30.08.2026

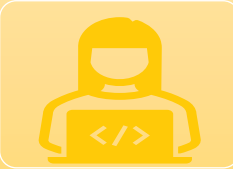
Learning objectives



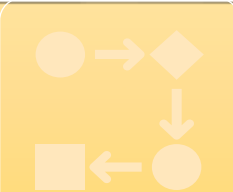
Concepts of single cell RNA-seq biological insights



Cell proportion changes

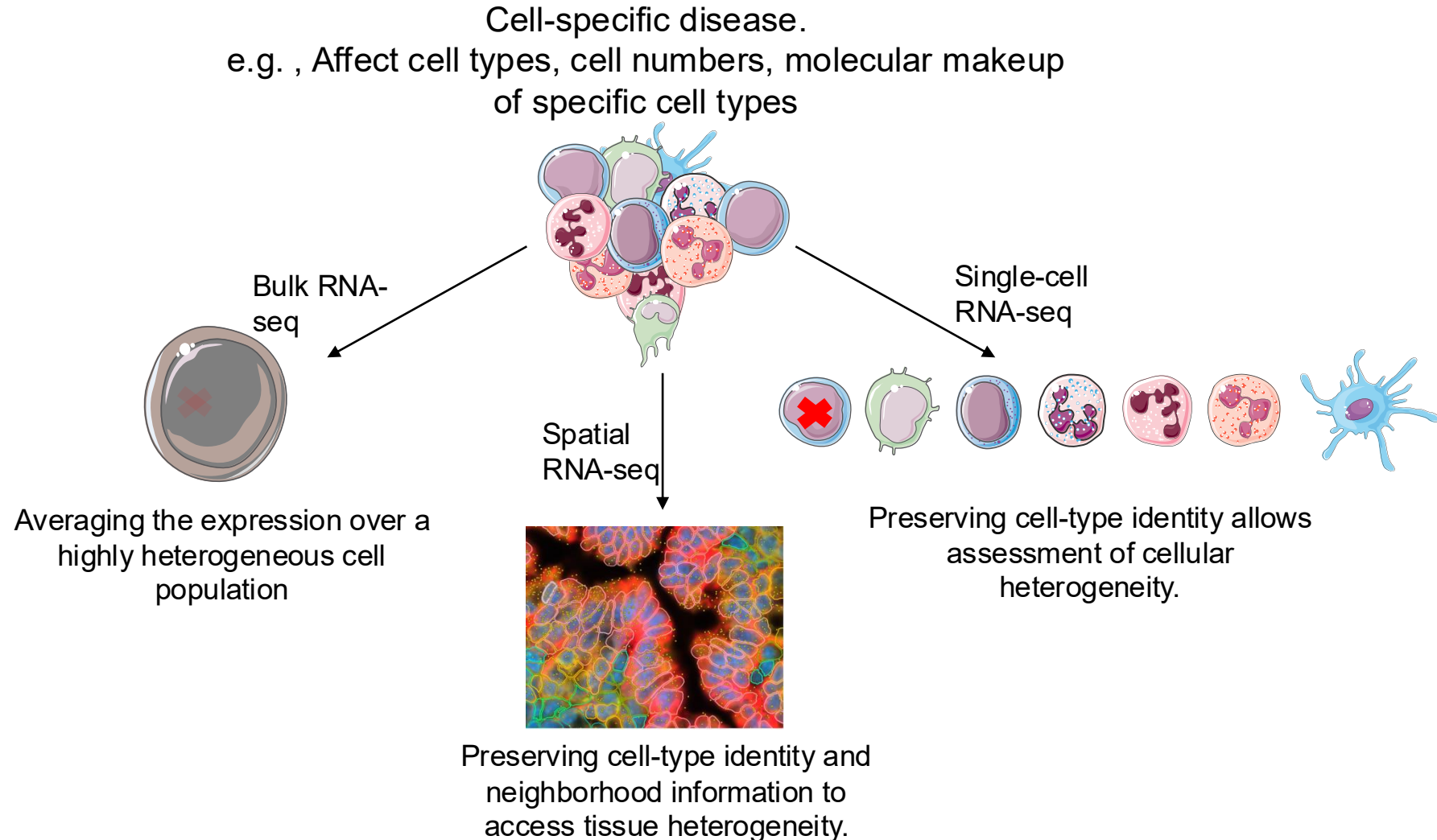


Differential expression analysis and enrichment



Other useful packages

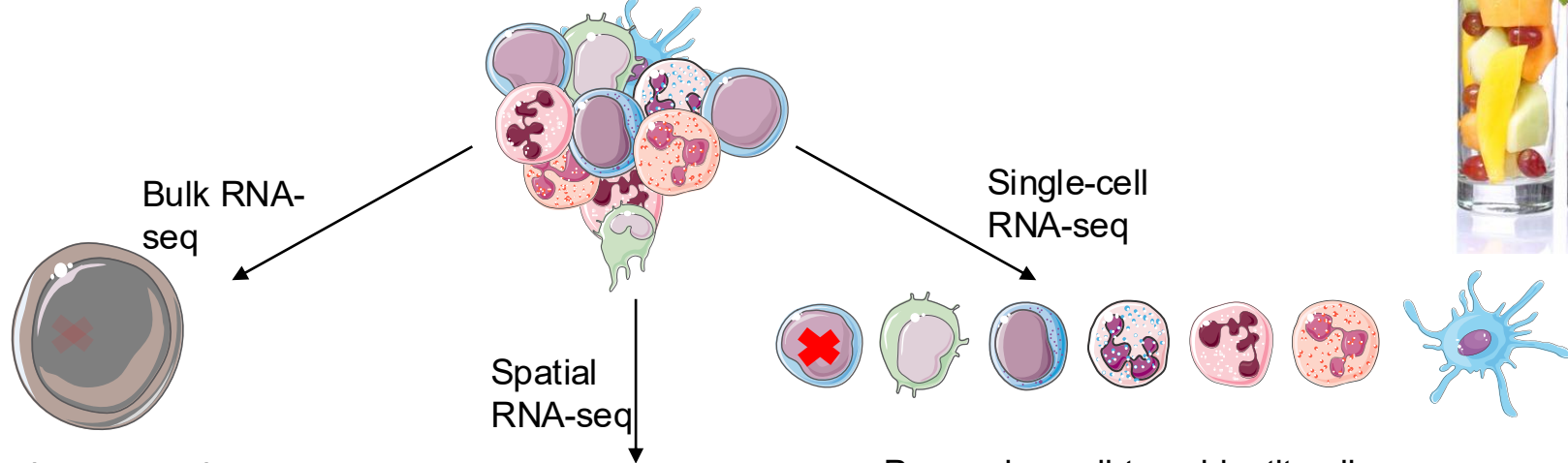
Transcriptome analysis for diseases



Something is rotten in my smoothie...



Cell-specific disease.
e.g. , Affect cell types, cell numbers, molecular makeup
of specific cell types



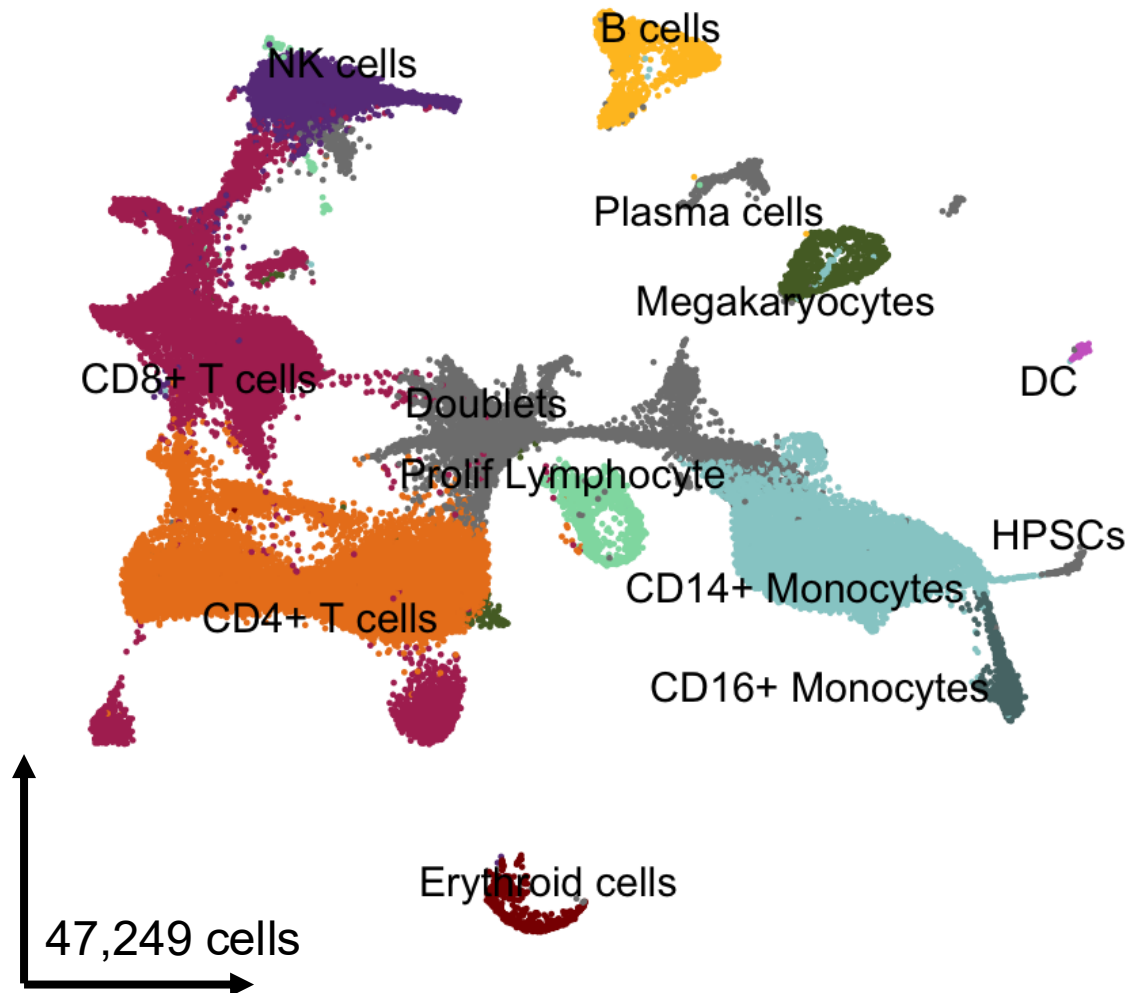
Averaging the expression over a
highly heterogeneous cell
population

Preserving cell-type identity allows
assessment of cellular
heterogeneity.

Preserving cell-type identity and
neighborhood information to
access tissue heterogeneity.



Your object



- B cells
- CD14+ Monocytes
- CD16+ Monocytes
- CD4+ T cells
- CD8+ T cells
- DC
- Erythroid cells
- Megakaryocytes
- NK cells
- Prolif Lymphocyte

**Integrated
object**

- QC clusters
- cluster annotation

Aims of our class:

- Does COVID-19 change X , on average over both sexes?
 - main effect of disease (adjusted for sex) ``~ sex + disease``

Aims of our class:

- Does COVID-19 change X , on average over both sexes?
 - main effect of disease (adjusted for sex) ``~ sex + disease``
- Do female and male samples differ in X ?
 - main effect of sex (adjusted for disease) ``~ disease + sex``

Aims of our class:

- Does COVID-19 change X , on average over both sexes?
 - main effect of disease (adjusted for sex) $\sim \text{sex} + \text{disease}$
- Do female and male samples differ in X ?
 - main effect of sex (adjusted for disease) $\sim \text{disease} + \text{sex}$
- Is the effect of COVID-19 different in females and males?
 - sex $\times$ disease interaction $\sim \text{sex} * \text{disease}$

Discussion of sex in scRNAseq:

1. The unit of replication is the individual

5,000 cells from one woman are still one woman. Treating cells as replicates (*pseudoreplication*) gives tiny, misleading p-values.

2. Sex-stratified results are useful, but they do not prove a sex difference.

"Significant in females, not significant in males" is not a test of a sex difference (Gelman & Stern, 2006).

3. Sex-chromosome genes always differ between the sexes.

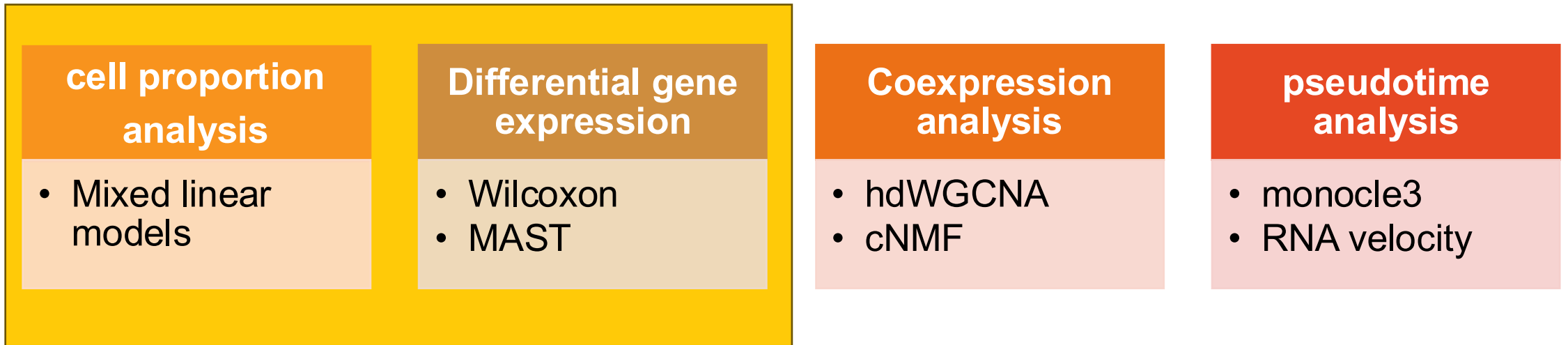
They can hide, or fake, other results if the design is unbalanced.

4. Power.

With one or two donors per group, a non-significant result tells us very little. We are exploring here, not making claims about COVID-19 biology.

Downstream analysis

- Dissecting the biological signals:



e.g. healthy vs. COVID or female COVID vs. male COVID

Cell proportion analysis

- Identifies significant changes in cell proportions based on treatment or other parameters.

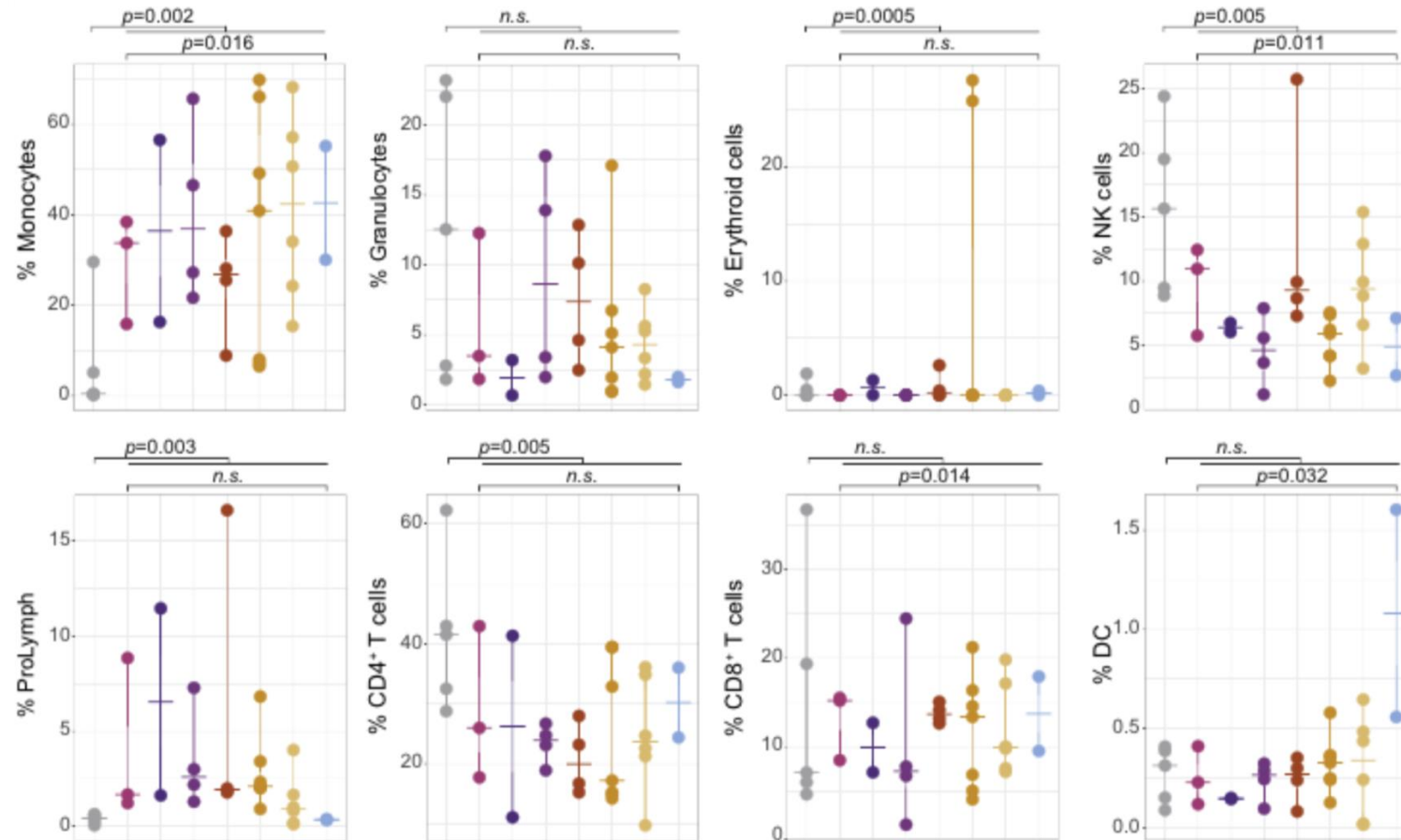
Disadvantages:

2. cell proportion analysis

- Mixed linear models

Cell proportion change through time: Mixed linear model

Cell proportion change between healthy and covid19: ANOVA



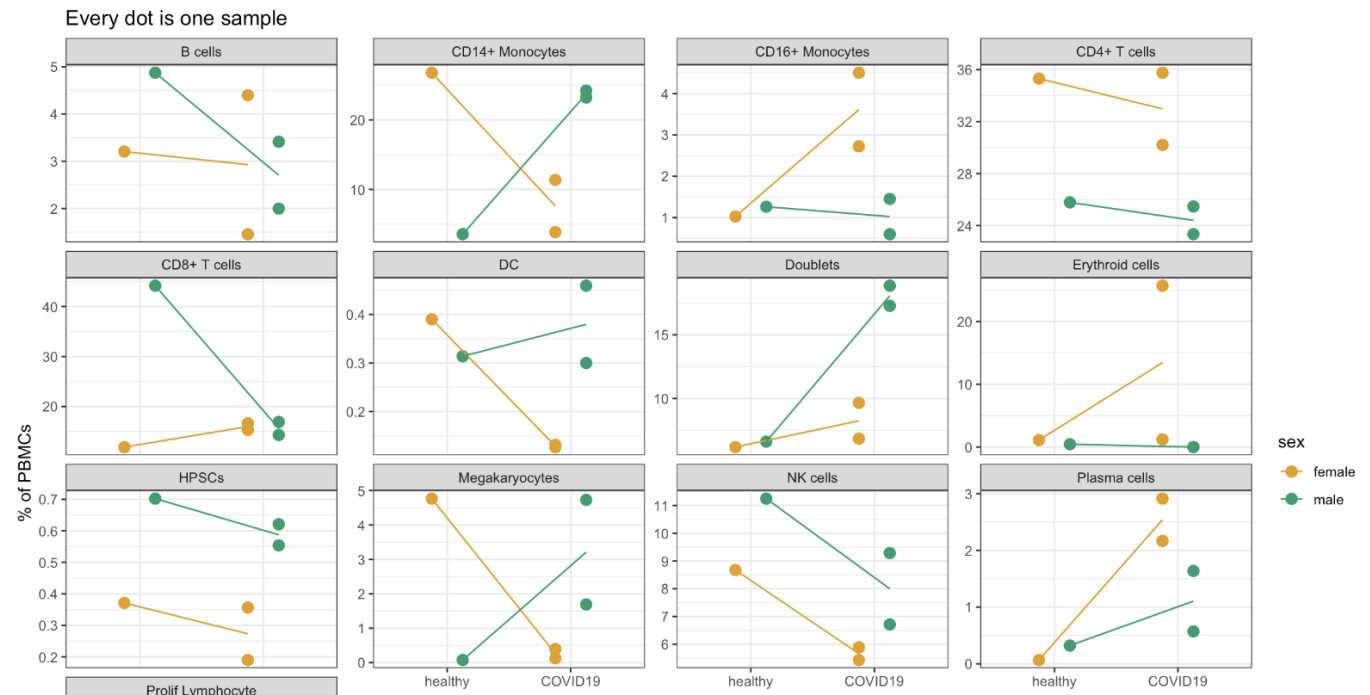
Cell proportion analysis: Exercise 2

- Visualization
- Propeller (Phipson et al., 2022, Bioinformatics) application of LMM in single-cell data
- Group per sample!

- What would be your cell type of interest?

Cell proportion analysis: Exercise 2

- Visualization
- Propeller (Phipson et al., 2022, Bioinformatics) application of LMM in single-cell data
- Group per sample!



- What would be your cell type of interest?

- Wilcox
- MAST

Differential expression analysis

- Identifies expression changes between treatments, for individual genes within a cell type.

Differential expression can be based on:

- Signature genes: compare one group against all other groups/clusters.
- Pairwise comparisons: compare one group (e.g. female vs. male) to another group.
- Longitudinal comparison: compare patient through time

Test	Model	Covariates	Notes
Wilcox	Wilcoxon rank-sum	No	Fast, especially with presto, which Seurat v5 uses automatically.
bimod	Likelihood ratio for bimodal expression	No	Historical option
roc	AUC classifier power	No	Returns "predictive power" instead of p-values
t	Student's t-test	No	Simple, assumes normality
LR	Logistic regression predicting group from expression	Yes	Lets you regress out batch or other factors
negbinom	GLM on counts	Yes	Only suitable for UMI data
MAST	Hurdle model	Yes	Seurat's wrapper only allows fixed effects, not random effects.
DESeq2	NB GLM	No	Meant for pseudobulk .

Differential expression: Exercise 3

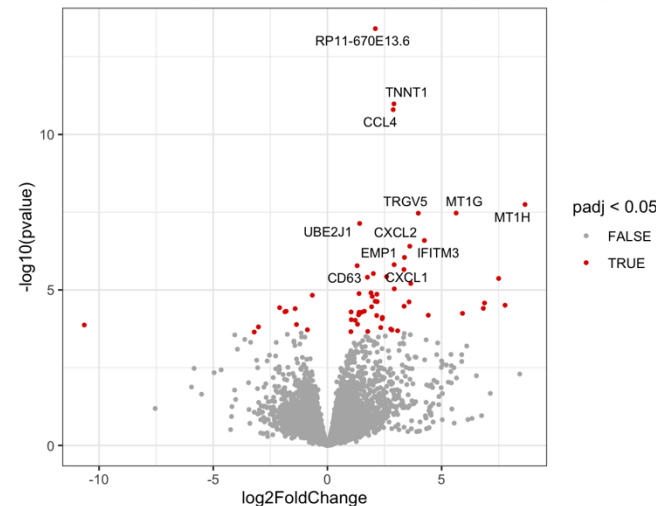
- Calculation per cell-type and matching with sex-linked genes
- Pseudobulk analysis and matching with sex-linked genes
- Adjusted models:
 - For sex
 - For disease

Differential expression: Exercise 3

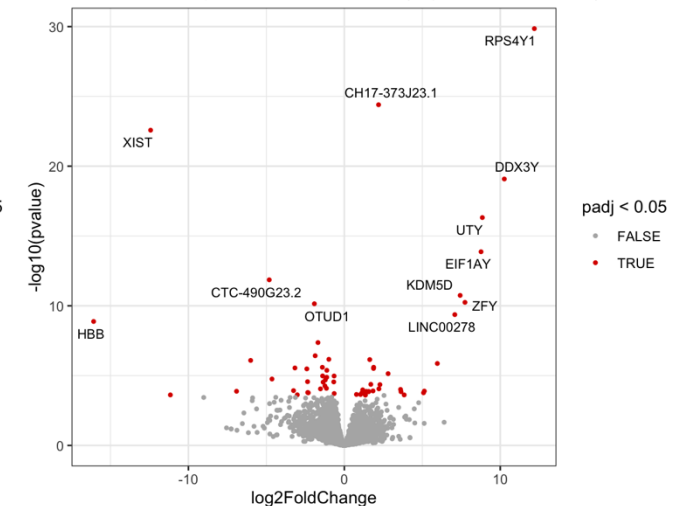
- Calculation per cell-type and matching with sex-linked genes
- Pseudobulk analysis and matching with sex-linked genes

- Adjusted models:
 - For sex
 - For disease

CD14+ Monocytes - COVID-19 vs healthy (adjusted for sex)



CD14+ Monocytes - male vs female (adjusted for disease)

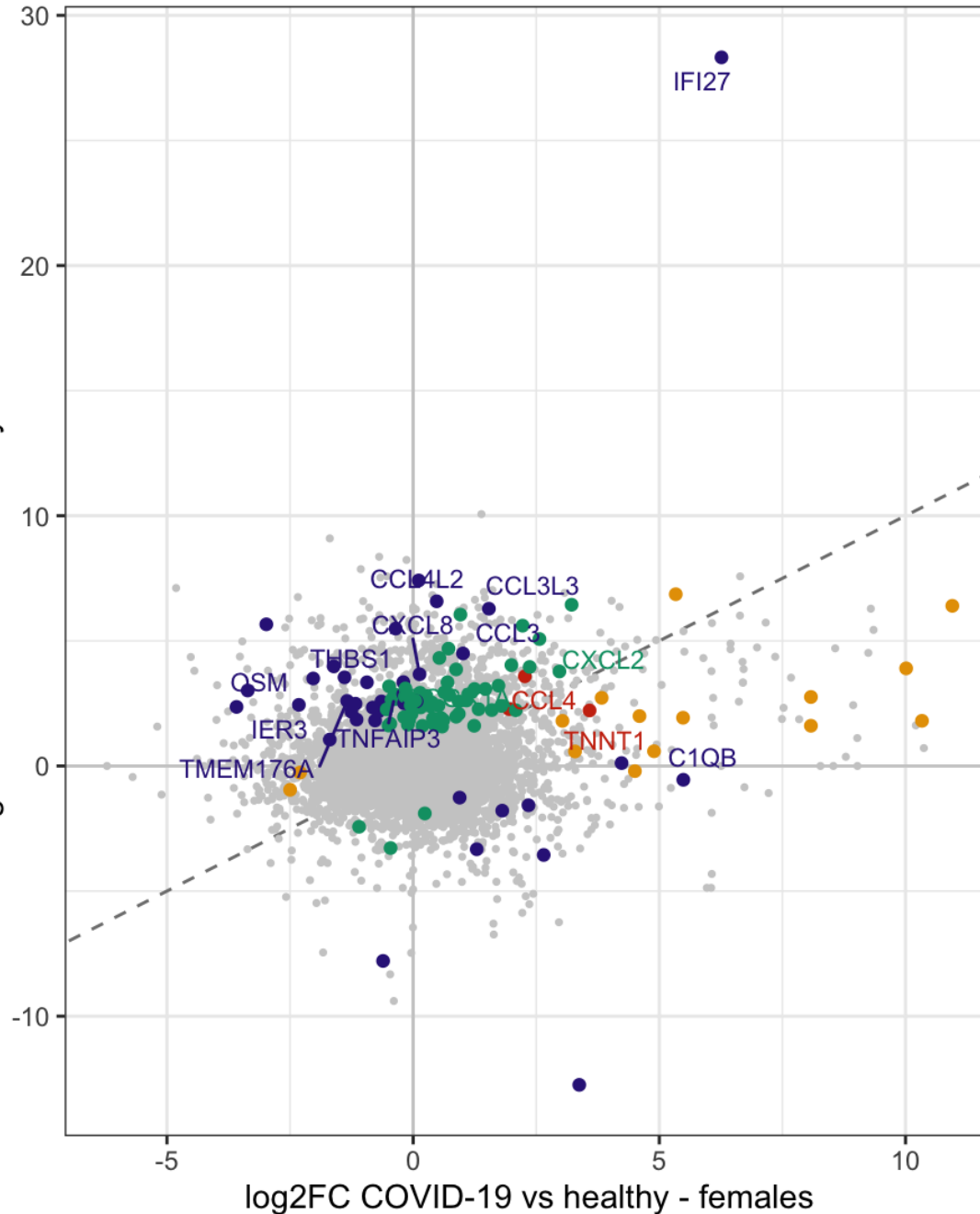


CD14+ Monocytes - COVID-19 response by sex

Differ

- Calculated
- Pseudocounts
- Adjusted for
 - For sex
 - For cell type
 - With

log2FC COVID-19 vs healthy - males



genes

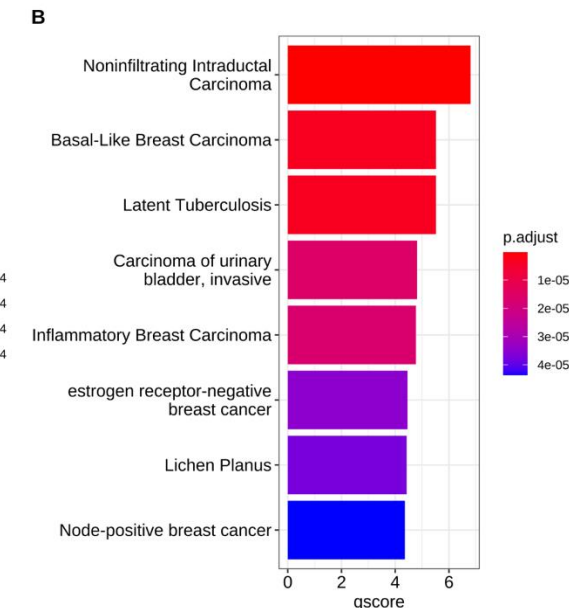
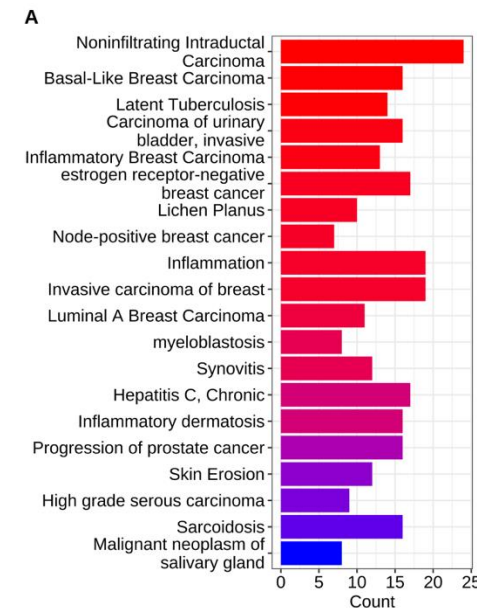
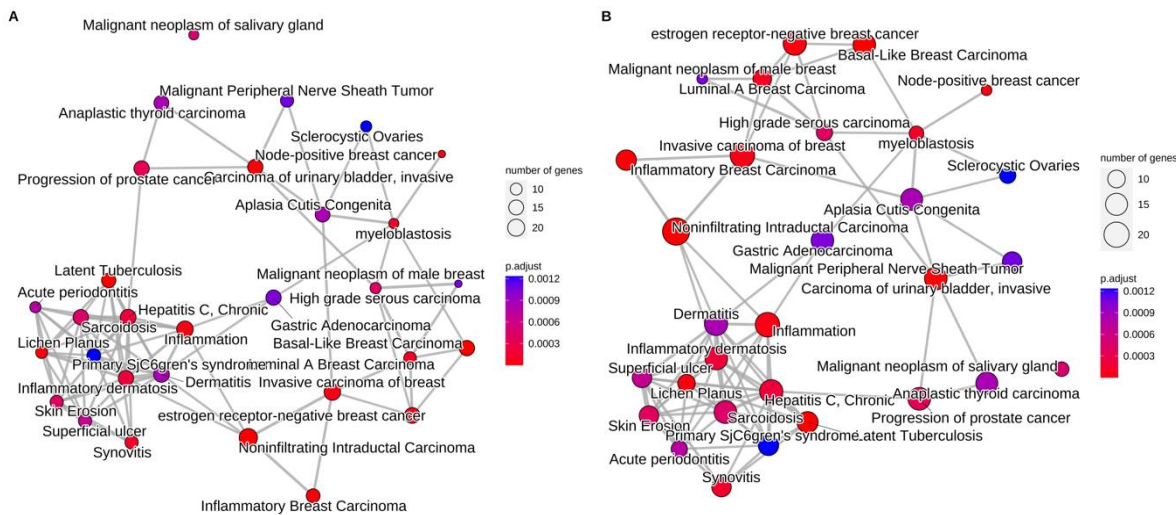
3

Enrichment analysis:

Determine whether certain biological terms or features are overrepresented or underrepresented in a dataset compared to what would be expected by chance.

3. DE: enrichment analysis

- Over-representation
- GSEA



Enrichment analysis types:

- Overrepresentation analysis (ORA):

A predefined set of genes or biological terms (e.g., pathways, gene ontology terms) is overrepresented in a list of genes of interest compared to a background set.

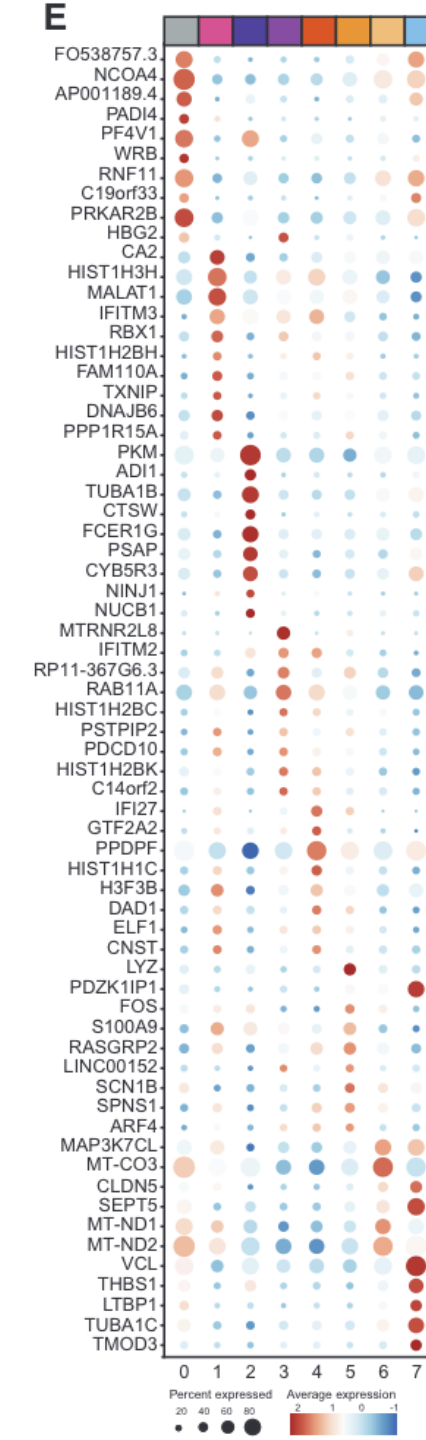
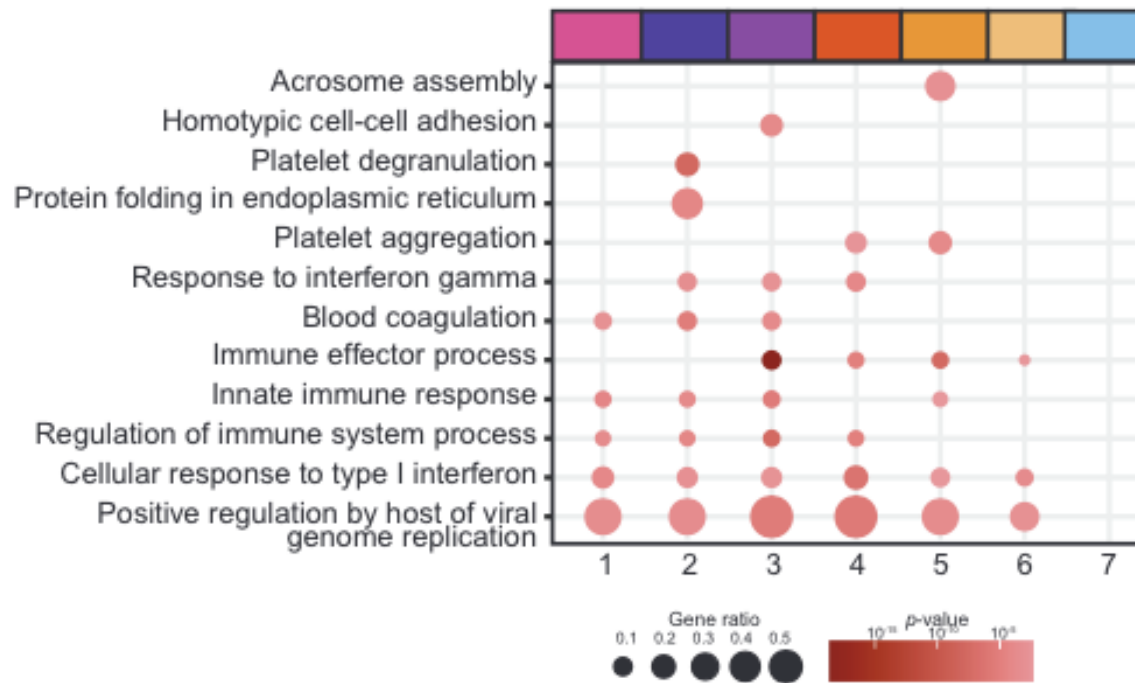
- hypergeometric test, Fisher's exact test, or chi-square

- Gene Set Enrichment Analysis (GSEA)

A predefined gene sets (e.g., pathways, gene ontology terms) show statistically significant differences in expression or other omics measures: e.g., ranked expression between different experimental conditions or phenotypes.

- nonparametric statistical test: Kolmogorov-Smirnov test.

Differentially expressed genes can also be used to inform GO terms enriched throughout timepoints

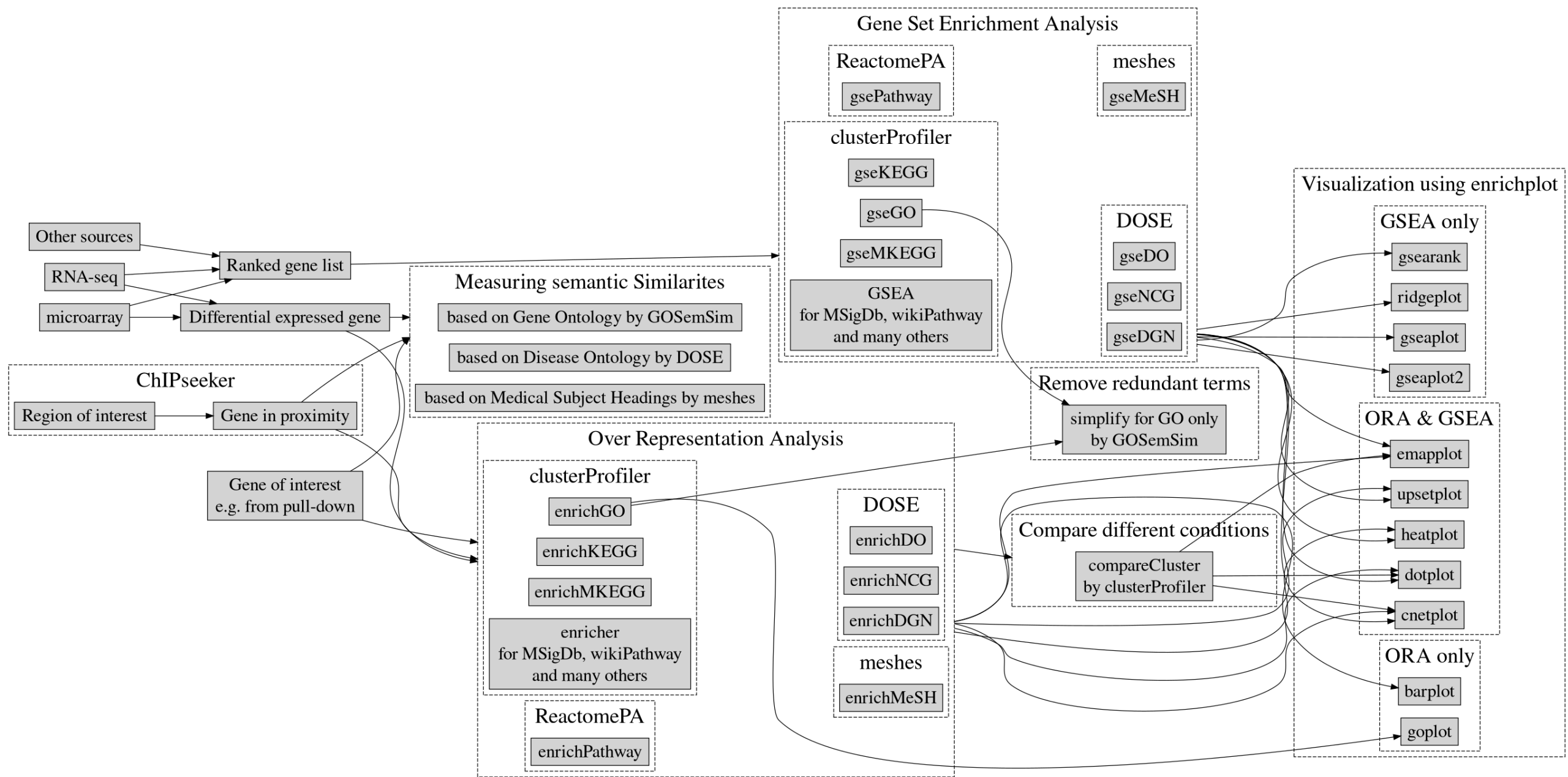


Tips

Collect the the enriched input: which genes/metabolites/...
enriched for each term.

Set boundaries to your enrichment analysis. You cannot do them
all (most of the time..)

If you cannot interpret the biology, ask someone.

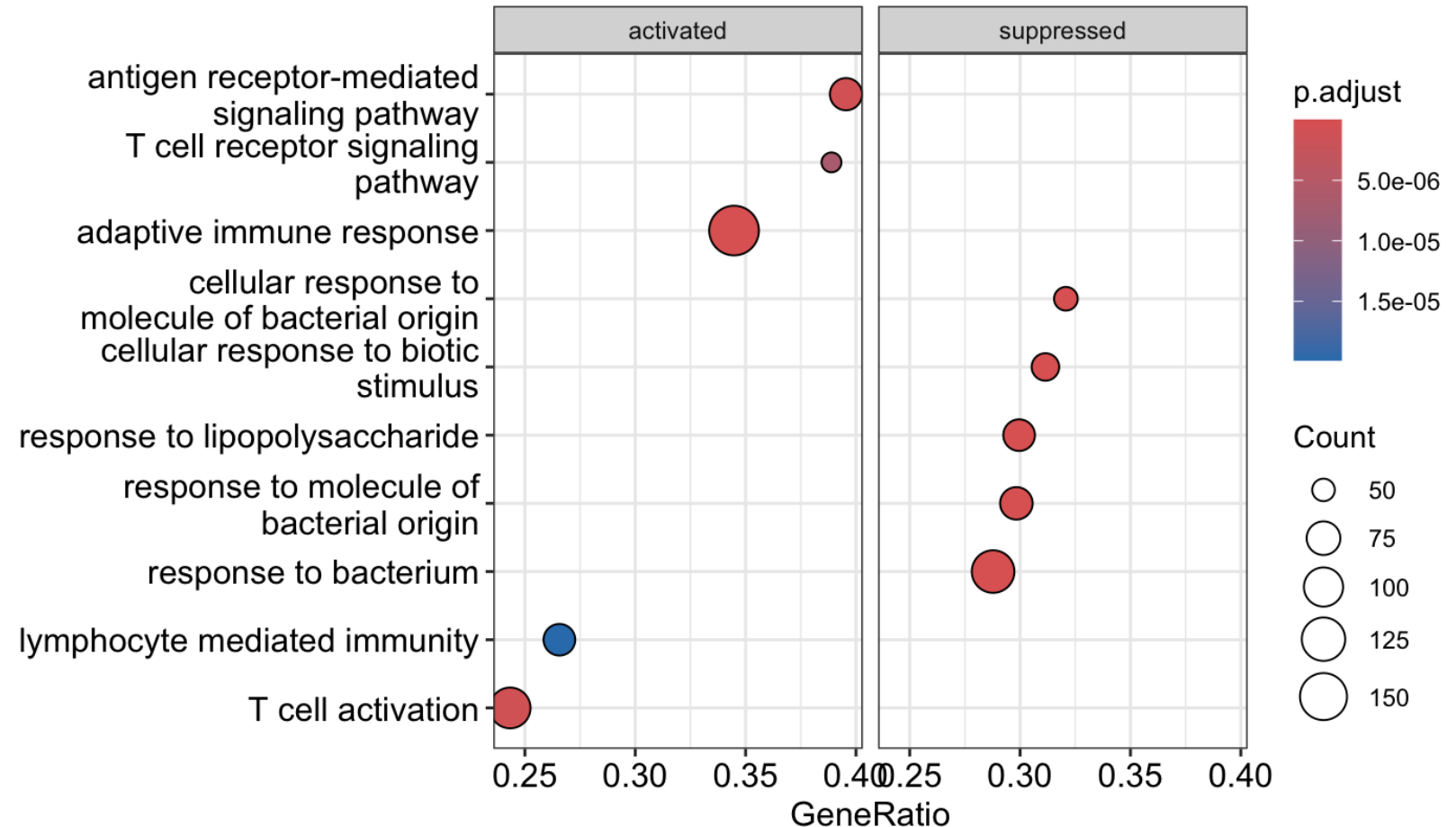


Enrichment analysis: Exercise 4

- ORA
- GSEA

Enrichment analysis: Exercise 4

- ORA
- GSEA



Where can I get public data?

- www.fastgenomics.org
- <https://www.ncbi.nlm.nih.gov/geo/>
- <https://support.10xgenomics.com/single-cell-gene-expression/datasets/>
- https://portals.broadinstitute.org/single_cell
- <https://preview.data.humancellatlas.org/>
- <http://jinglebells.bgu.ac.il/>
-

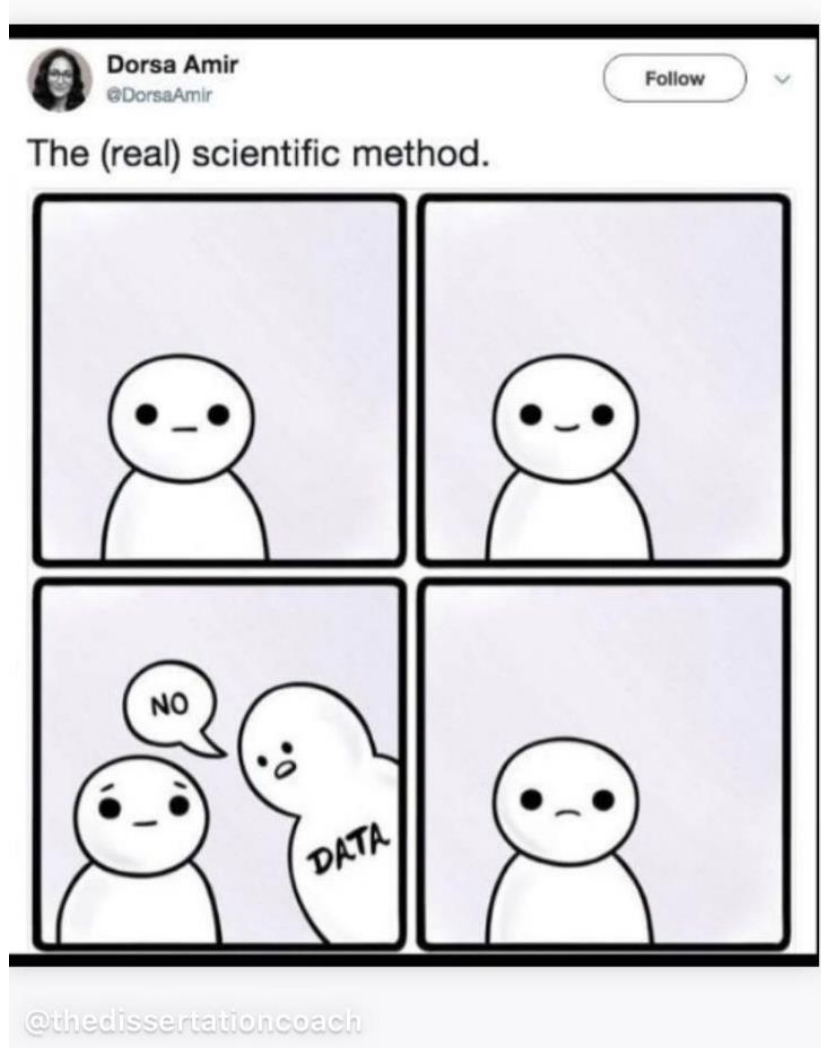
FASTGENOMICS



10X GENOMICS



Questions?



Hypothesis-driven modus operandi

